

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Data Interpretation (Medium)

Weightage: 20%

QUESTIONS: Total Marks: 20

1: HTTP Response Analysis

A web server returns the following HTTP response header:

```
HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: -150
Connection: keep-alive
```

Questions:

1. Identify any incorrect or invalid fields in the header.
2. Analyze the issue with `Content-Length` and explain its impact.
3. Determine whether the status code matches the response correctness.
4. Suggest corrections for the identified errors.

2: DOM Structure Analysis

Consider the following DOM tree representation:

```
<html>
<head>
<title>TestPage</title>
</head>
```

```
<body>
<div>
<p>HelloWorld
</div>
</body>
</html>
```

Questions:

1. Identify missing or improperly closed elements.
2. Analyze how the browser will interpret this structure.
3. Determine the impact on rendering and layout.
4. Suggest corrections to make the DOM valid.

3: GET vs POST in Login System

A login system uses two different methods:

Method	URL Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL	Low
POST	/login	Hidden in body	Moderate

Questions:

1. Interpret the differences in data transmission between GET and POST.
2. Analyze which method is more suitable for login and why.
3. Identify security risks in using GET for authentication.
4. Suggest best practices for secure login implementation.

4: HTML Code Inspection

Analyze the following HTML snippet:

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome
<p>This is a paragraph

</body>
</html>
```

Questions:

1. Identify missing closing tags.

2. Analyze issues with the `<img>` tag.
3. Determine how browsers will handle these errors.
4. Suggest a corrected version of the code.

5: Browser-Server Communication Flow

The following sequence describes a web request:

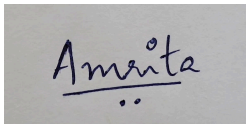
1. User enters a URL in the browser
2. DNS lookup occurs
3. HTTP request is sent
4. Server processes request
5. Response is returned
6. Browser renders page

Questions:

1. Interpret each step in the communication flow.
2. Identify where delays can occur.
3. Analyze the role of DNS in the process.
4. Suggest optimizations to improve performance.

Code of Conduct:

I AMRITA ANAIKUMAR - 192471035 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'Amrita' with a small mark below it.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

1: HTTP Response Analysis

A web server returns the following HTTP response header:

HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: -150
Connection: keep-alive

Questions:

1. Identify any incorrect or invalid fields in the header.
2. Analyze the issue with Content-Length and explain its impact.
3. Determine whether the status code matches the response correctness.
4. Suggest corrections for the identified errors.

Given Header

HTTP/1.1 200 OK
Date: Mon, 01 Apr 2026 10:00:00 GMT
Content-Type: text/html
Content-Length: -150
Connection: keep-alive

1. Identify any incorrect or invalid fields in the header.

- **Content-Length: -150** is invalid because Content-Length cannot be negative.
- Other fields (HTTP/1.1, 200 OK, Date, Content-Type, and Connection) are valid.

2. Analyze the issue with Content-Length and explain its impact.

- Content-Length specifies the size of the response body in **bytes**.
- A negative value is invalid.
- It may cause:
 - Browser errors.
 - Incomplete page loading.
 - Failure to display content.
 - Connection or parsing errors.

3. Determine whether the status code matches the response correctness.

- **Status Code: 200 OK** means the request was processed successfully.
- Since the header contains an invalid Content-Length, the response is **not completely correct**.

- The status code does not match the actual response correctness.

4. Suggest corrections for the identified errors.

Correct header:

HTTP/1.1 200 OK

Date: Mon, 01 Apr 2026 10:00:00 GMT

Content-Type: text/html

Content-Length: 150

Connection: keep-alive

- Replace -150 with the correct positive size of the response body.

2: DOM Structure Analysis

Consider the following DOM tree representation:

```
<html>
<head>
<title>TestPage</title>
</head>
<body>
<div>
<p>HelloWorld
</div>
</body>
</html>
```

Questions:

1. Identify missing or improperly closed elements.
2. Analyze how the browser will interpret this structure.
3. Determine the impact on rendering and layout.
4. Suggest corrections to make the DOM valid.

ANSWER:

1. Identify missing or improperly closed elements.

- Missing `</p>` tag.
- Missing `</div>` was present correctly, but the paragraph should be closed before it.
- Missing `<!DOCTYPE html>`.

2. Analyze how the browser will interpret this structure.

- Browsers automatically insert the missing `</p>` tag.

- The page usually displays correctly.
- Different browsers may handle the correction differently.

3. Determine the impact on rendering and layout.

- The page may still display normally.
- Invalid HTML can cause:
 - Inconsistent rendering.
 - CSS layout problems.
 - DOM validation errors.

4. Suggest corrections to make the DOM valid.

```
<!DOCTYPE html>
<html>
<head>
<title>TestPage</title>
</head>
<body>
<div>
<p>Hello World</p>
</div>
</body>
</html>
```

3: GET vs POST in Login System

A login system uses two different methods:

Method	URL	Example	Data Visibility	Security Level
GET	/login?user=admin&pass=1234	Visible in URL	Low	
POST	/login	Hidden in body	Moderate	

Questions:

1. Interpret the differences in data transmission between GET and POST.
2. Analyze which method is more suitable for login and why.
3. Identify security risks in using GET for authentication.
4. Suggest best practices for secure login implementation.

ANSWER:

1. Interpret the differences in data transmission between GET and POST.

GET	POST
Sends data in the URL	Sends data in the request body
Data is visible	Data is hidden from the URL
Used for retrieving data	Used for submitting data
Limited data size	Can send larger amounts of data

2. Analyze which method is more suitable for login and why.

POST is more suitable because:

- Credentials are sent in the request body.
- Username and password are not exposed in the URL.
- It provides better security than GET.

3. Identify security risks in using GET for authentication.

- Username and password appear in the URL.
- URLs may be stored in browser history.
- URLs may be logged by servers.
- URLs can be bookmarked or shared accidentally.
- Sensitive information may be exposed.

4. Suggest best practices for secure login implementation.

- Use **POST** instead of GET.
- Use **HTTPS** to encrypt communication.
- Validate input on both client and server.
- Hash passwords before storing them.
- Implement secure session management.
- Use strong authentication and authorization.

4: HTML Code Inspection

Analyze the following HTML snippet:

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome
<p>This is a paragraph

</body>
</html>
```

Questions:

1. Identify missing closing tags.
2. Analyze issues with the tag.
3. Determine how browsers will handle these errors.
4. Suggest a corrected version of the code.

ANSWER:

1. Identify missing closing tags.

Missing:

- </h1>
- </p>

2. Analyze issues with the tag.

- Missing the alt attribute.
- alt text improves accessibility.
- Example:

```

```

3. Determine how browsers will handle these errors.

Browsers generally:

- Automatically close missing tags.
- Render the page as best as possible.
- Different browsers may produce slightly different layouts.

4. Suggest a corrected version of the code.

```
<!DOCTYPE html>
<html>
<head>
<title>Sample</title>
</head>
<body>
<h1>Welcome</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

5: Browser-Server Communication Flow

The following sequence describes a web request:

- 1. User enters a URL in the browser
- 2. DNS lookup occurs
- 3. HTTP request is sent
- 4. Server processes request
- 5. Response is returned
- 6. Browser renders page

Questions:

- 1. Interpret each step in the communication flow.
- 2. Identify where delays can occur.
- 3. Analyze the role of DNS in the process.
- 4. Suggest optimizations to improve performance.

ANSWER:

1. Interpret each step in the communication flow.

Step	Description
1. User enters URL	User types the website address in the browser.
2. DNS Lookup	DNS converts the domain name into an IP address.
3. HTTP Request	Browser sends an HTTP request to the web server.
4. Server Processing	Server processes the request and prepares the response.
5. HTTP Response	Server sends the requested webpage or data back to the browser.

6. Browser Rendering	Browser interprets HTML, CSS, and JavaScript and displays the webpage.
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2. Identify where delays can occur.

Delays may occur during:

- DNS lookup.
- Slow internet connection.
- Server processing.
- Large response size.
- Browser rendering of large pages.

3. Analyze the role of DNS in the process.

DNS (Domain Name System):

- Translates the domain name into an IP address.
- Allows the browser to locate the correct server.
- Without DNS, the browser cannot connect to the website.

4. Suggest optimizations to improve performance.

- Enable browser caching.
- Use Content Delivery Networks (CDNs).
- Compress files using Gzip or Brotli.
- Optimize images.
- Minify HTML, CSS, and JavaScript.
- Reduce HTTP requests.
- Use faster DNS servers.
- Improve server performance.